Behavioral Clustering of NBA Prediction Markets on Kalshi

Math 364 — Machine Learning in Asset Pricing

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What is Kalshi?

A federally regulated prediction market exchange

Kalshi is the first CFTC-regulated exchange where retail investors can trade binary outcome contracts on real-world events.

How contracts work

- Each contract is a YES/NO question on a future event
- Resolves to \$1.00 if the stated condition is met (YES)
- Resolves to \$0.00 if not (NO)
- Market price at any moment $\approx$ crowd's **implied probability** of YES

Example contract

"Will the Boston Celtics win Game 4?"

Price	Interpretation
\$0.30	30% implied prob. of YES
\$0.70	70% implied prob. of YES
\$0.99	Near-certain YES
\$0.01	Near-certain NO

As news arrives (injury reports, tip-off, live score), the price updates continuously, tracing a **probability path** from open to resolution.

Research Questions

Primary question

Do NBA prediction markets on Kalshi cluster into **distinct behavioral types** based on pre-resolution price dynamics?

Secondary questions

Q2 Do those clusters **align with** Kalshi's stated market structure (game winner, player prop, total points, etc.), or do they **cut across** it?

Does the behavioral taxonomy match the contractual taxonomy?

Q3 Are some clusters **systematically mispriced** relative to actual resolution outcomes?

Does the crowd consistently over- or under-estimate the probability of YES for certain behavioral types?

Core insight: if price path shape drives cluster membership, then behavioral dynamics reveal market structure that category labels alone cannot.

Data

Collection

- **Source:** Kalshi REST API v2 (public)
- **Period:** February – May 2026
- **Markets pulled:** 19,406 resolved NBA contracts across 24 series
- **After filtering** (duration $\geq$24 h, $\geq$10 candles): **946 markets**
- **Price history:** hourly OHLCV candlesticks per market (49,015 records total)

Market types in final sample

Type	n
Half	124

Engineered features (per market)

Feature	Description
`final_prob`	YES prob. in final hour
`change_3h`	Price move, last 3 h
`change_24h`	Price move, last 24 h
`total_range`	Max−min over lifetime
`volatility`	Std. dev. of hourly Δp
`drift_slope`	OLS slope, final 24 h
`days_open`	Market lifetime (days)
`time_near_50`	Fraction of life near 50%

Features capture the *shape* of each market's probability path, not its outcome level. All 8 features are z-scored before clustering.

Methodology

K-means clustering

1. Represent each market as an 8-dimensional feature vector (z-scored)
2. K-means partitions markets into k groups by minimizing within-cluster variance
3. Cluster $\approx$ "markets with similar price path behavior"
4. 10 random restarts; best solution kept

Choosing k

- **Elbow method:** plot inertia (within-cluster SSE) vs. k; look for the bend where gains diminish

What features capture behaviorally

Feature	Behavioral meaning
`final_prob`	Was the market priced as a favorite or underdog?
`change_24h`	Did the crowd update strongly in the final day?
`total_range`	How uncertain was the outcome at any point?
`volatility`	How rough was the price path?
`time_near_50`	How long did the market sit in genuine uncertainty?
`days_open`	Was this a long-horizon or short-horizon market?

Calibration test

Selecting the Number of Clusters

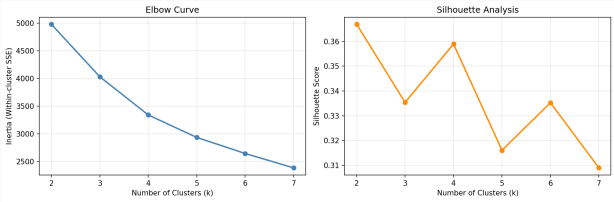


Figure 1: Inertia (left) and silhouette score (right) for $k = 2$–7.

Reading the charts

- The elbow in inertia occurs at $k = 3$–4, after which gains diminish
- Silhouette peaks globally at $k = 2$ (trivially separates YES-side from NO-side markets)
- Among $k \geq 4$, $k = 4$ achieves the highest silhouette score of **0.36**

Selected: $k = 4$

Four clusters yield meaningful behavioral distinctions while maintaining good cluster cohesion

Cluster Scatter Plot (PCA)

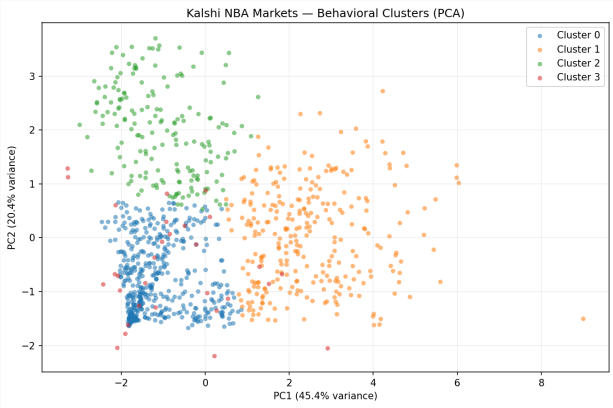


Figure 2: All 946 markets projected onto the first two PCA components, colored by cluster assignment.

Interpreting the scatter

Each point is one resolved NBA market. Axes are the first two principal components of the 8-dimensional feature space, capturing the dominant directions of behavioral variation.

- **PC1** (horizontal) primarily separates markets by `final_prob` — how the market resolved
- **PC2** (vertical) captures path dynamics: range, volatility, time near 50%

Cluster Profiles: What Each Cluster Behaves Like

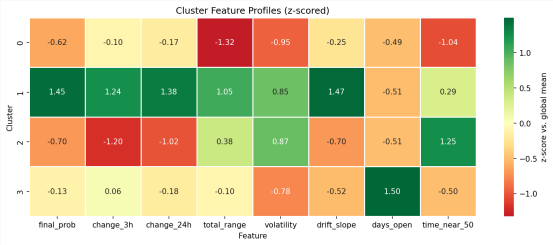


Figure 3: Mean feature value per cluster (z-scored across clusters). Red = above average; green = below average.

Cluster 0 — "Stable Underdogs" ($n = 453$)

Consistently priced <25%, almost never near 50%, low volatility. Outcome was obvious from market open. Resolved NO 86% of the time.

Cluster 1 — "Late Movers" ($n = 286$)

Started near 50/50, then surged +48 cents in the final 24 hours. Wide range (0.73). Resolved YES 84% of the time — the crowd moved decisively and correctly.

Cluster 2 — "Volatile Near-Misses" ($n = 177$)

Spent 46% of lifetime near 50%, then collapsed toward NO. Wide range (0.59), strong negative drift. Markets that *looked* contested but resolved NO.

Do Clusters Align with Market Type?

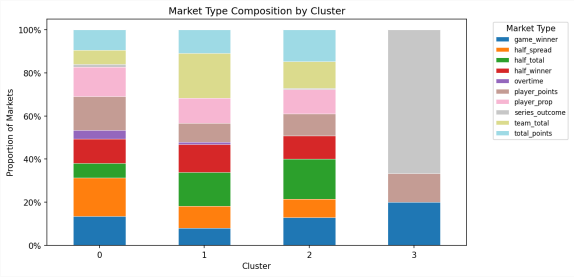


Figure 4: Market type composition within each cluster (row-normalized to 100%).

Key finding: clusters cut across market type

Every cluster contains both game-outcome markets and player prop markets. The behavioral taxonomy does **not** mirror Kalshi's stated categories.

What this means

- A game winner market and a player prop market can share the same price-path behavior — they end up in the same cluster
- The relevant dimension is **how the crowd updates beliefs**, not what the contract is *about*

Mispricing Analysis

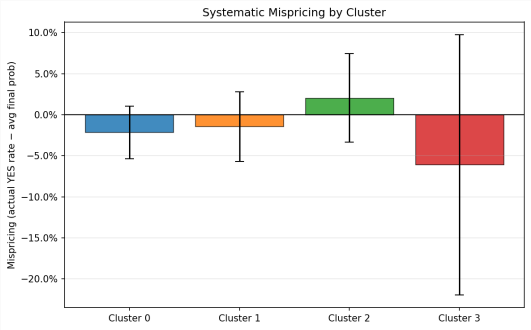


Figure 5: Mispricing per cluster (actual YES rate − avg. final prob.) with 95% Wald CIs. No asterisk = not significant at $\alpha = 0.05$.

Results

Cluster	n	Avg. prob	YES rate	Mispricing
0	453	16.3%	14.1%	−2.1 pp
1	286	85.4%	83.9%	−1.5 pp
2	177	13.8%	15.8%	+2.1 pp
3	30	32.8%	26.7%	−6.1 pp

All p-values > 0.22; no cluster is significantly mispriced at $\alpha = 0.05$.

Interpretation

- **Clusters 0 & 1** (stable underdogs & late movers): slightly over-priced relative to outcomes — markets slightly overestimate YES
- **Cluster 2** (volatile near misses): slightly

Calibration Curves

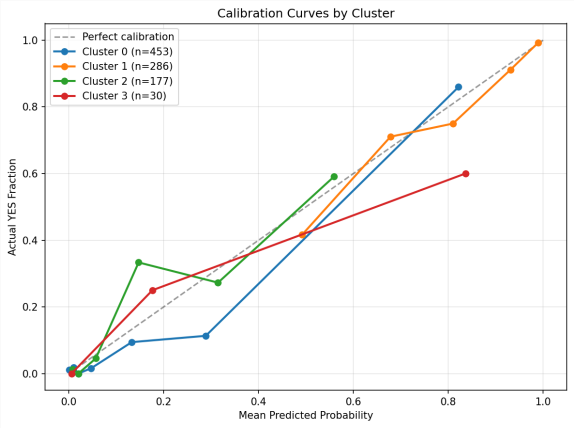


Figure 6: Quantile-binned calibration curves per cluster. A market that lies on the dashed diagonal is perfectly calibrated.

What calibration means

A market is **well-calibrated** if, among all contracts priced at p, a fraction p actually resolve YES. Perfect calibration = the diagonal.

Good calibration is a necessary (but not sufficient) condition for market efficiency.

Findings by cluster

- **Clusters 0 & 1** hug the diagonal closely at their respective ends of the probability range — these markets are highly calibrated

Conclusions

Q1: Do distinct behavioral types exist?

Yes. K-means finds 4 well-separated clusters (silhouette = 0.36) in the 8-dimensional feature space. The four types are: Stable Underdogs, Late Movers, Volatile Near-Misses, and Long-Horizon Markets.

Q2: Align with or cut across Kalshi's structure?

Cut across. Every cluster mixes game-winner markets with player prop markets. Behavioral dynamics — not contract category — are the true organizing principle. The exception is Cluster 3, where duration alone drives separation.

What the clusters reveal

The primary axes of NBA prediction market behavior are:

1. **Where a market resolves:** near 0 (underdog) or near 1 (favorite) — the dominant dimension
2. **How it gets there:** steadily priced from open (Cluster 0) vs. sharp late convergence (Cluster 1) vs. contested-then-collapsed (Cluster 2)
3. **How long it was open:** short-term game-day contracts vs. multi-week series markets (Cluster 3)

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